



USAT Project Podium

Metabolic Testing and Data Analysis Update

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Project Podium Overview



Strength	Description
Daily Training Environment	- Immersive, year-round environment integrating swim, bike, and run sessions with strength, recovery, and tactical skills work. - Small, focused athlete group allows for individualized programming and daily coach–athlete interaction. - Emphasis on fostering a competitive yet collaborative training culture that prepares athletes for the demands of the World Triathlon circuit.
Supreme Supporting Partners	- Backed by world-class equipment, apparel, and nutrition sponsors, ensuring athletes have the same tools as Olympic medal contenders. - Access to cutting-edge sport science through USOPC, university partners, and technology providers for performance testing, biomechanics, and data analysis. - Integrated support team including physiotherapists, nutritionists, sport psychologists, and bike mechanics. - Partnerships that offer mentorship, networking, gear, and professional development opportunities.
Scholarships and Support	- Partnership with Arizona State University provides tuition assistance, enabling athletes to pursue degrees while training and competing. - Academic advising and flexible class scheduling tailored to elite race calendars. - Life skills and career development workshops to prepare athletes for post-sport careers. - Proven track record of producing graduates who excel both in sport and in professional pathways.
Facilities and Resources	- Access to world-class training venues in Tempe, AZ, and Park City, UT: Elite Pools, private open water swimming areas, indoor/outdoor tracks, dedicated cycling routes, and altitude training opportunities. - On-site recovery and rehabilitation facilities, including ice baths, compression therapy, and sports massage. - High-tech performance lab for physiological testing, lactate analysis, and power/HR monitoring. - Year-round training in an optimal climate that supports consistent outdoor swimming, cycling, and running.

Competition Overview

- Project Podium athletes benefit from unique competitive opportunities spanning domestic, regional, and international levels, enabling them to gain experience, earn ranking points, and develop toward the world stage.

Domestic and Regional Competition



USAT National Championships – Project Podium athletes have excelled at the junior national level, winning the last 5 USA National Championships

Development Pipeline – Provides race opportunities for younger athletes to build skills, confidence, and readiness for international racing

World Triathlon Hosted Events



Continental Cups – Foundation of senior elite international racing, giving athletes the chance to earn valuable World Triathlon ranking points and gain high-level race experience

World Cups – Global-level elite races attracting strong international fields, offering higher points and exposure

World Triathlon Championship Series (WTCS) – Premier annual series featuring the world's top athletes, awarding maximum points outside the Olympics

Supertri



Stars and Stripes Racing – Professional short-course triathlon team coached by Project Podium Head Coach Parker Spencer, providing athletes with a supportive, high-performance team environment

New Short-Course Competition – Innovative race formats that emphasize speed, tactical racing, and global exposure, giving athletes additional opportunities to compete against world-class fields.

Data Analysis Overview



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World Triathlon Championship Series (WTCS) – Premier annual series featuring the world's top athletes, awarding maximum points outside the Olympics

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Race Overview – WTCS Yokohama

WHO

- Three Project Podium Athletes competed at the WTCS Yokohama – Braxton Legg, Blake Bullard, Reese Vannerson
- These athletes have similar metabolic profiles; however historically Reese Vannerson has been the top runner



WHAT

- All three athletes came off the bike together in the main chase group. In this position, Reese was most likely to have the best performance
- Yet, he struggled on the run and finished 3rd among the group



HOW

- When analyzing the athletes bike data, Reese's file in terms of average power looks very similar to Blake's and Braxton's.
- However, using advanced simulation techniques we model out Reese's Blood Lactate levels and see that he finished the bike far above his threshold. This led to poor run performance



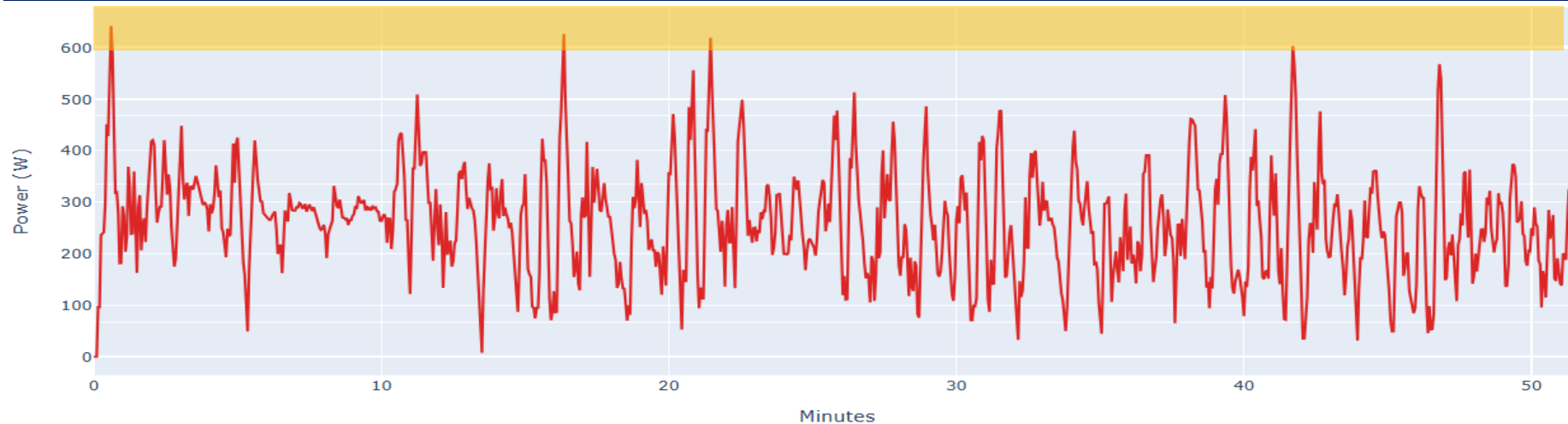
WHY

- This is important because without this data we might have believed Reese to have lost significant run fitness, whereas he spiked his lactate by surging over 600W several times due to pack positioning

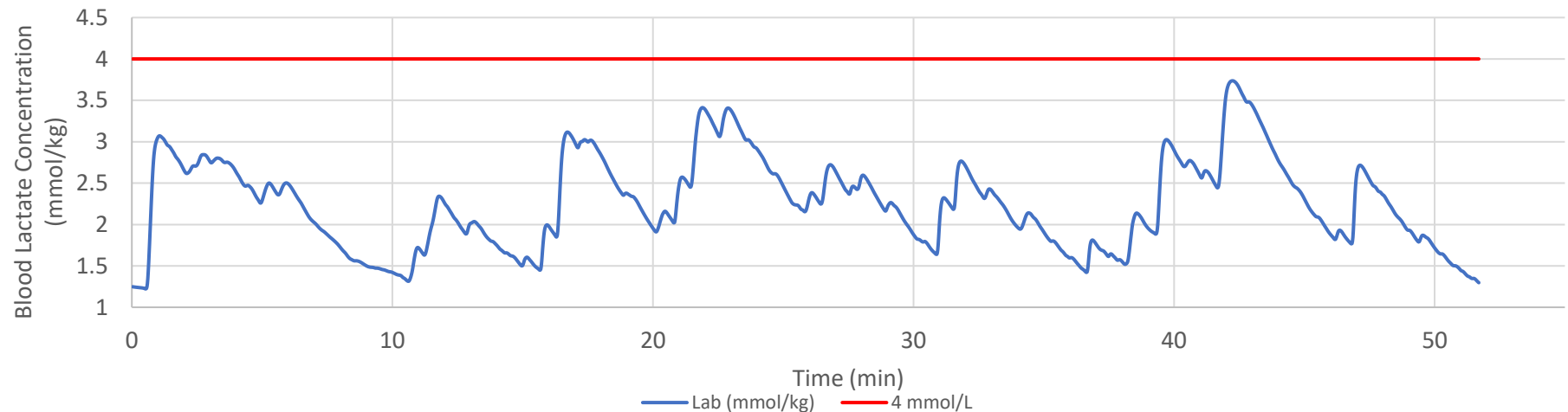
Blake Bullard Yokohama Bike

- Despite being in the breakaway, Blake minimized his surges over 600W once. Then when falling back into the main chase, Blake rode mostly around a medio effort keeping lactate below 4 mmol/kg

Raw Power File



Simulated Blood Lactate Concentration

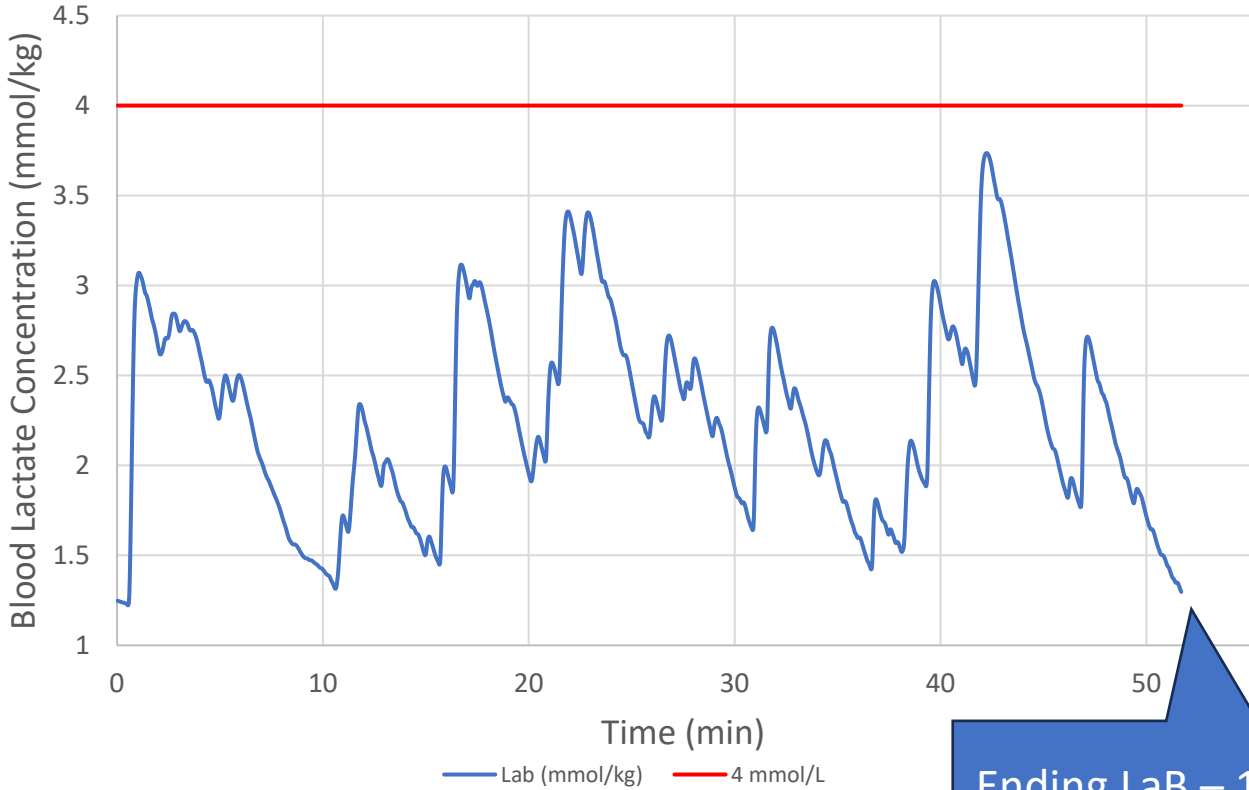


Note: Power file shown with 2s smoothing for better visibility

Blake Bullard Yokohama Bike

- Blake began the bike in a small 7-man breakaway coming out of the water yet had his highest spikes in lactate once he had been dropped and settled into the main chase group. This suggests that Blake needs to continue working on bike skills rather than pure power numbers
- Blake relatively even pacing resulted in a VI of 1.08, always keeping lactate below 4 mmol/kg and even dropped it all the way below 2 mmol/kg before starting the run

Simulated Blood Lactate Concentration



Ending LaB – 1.3

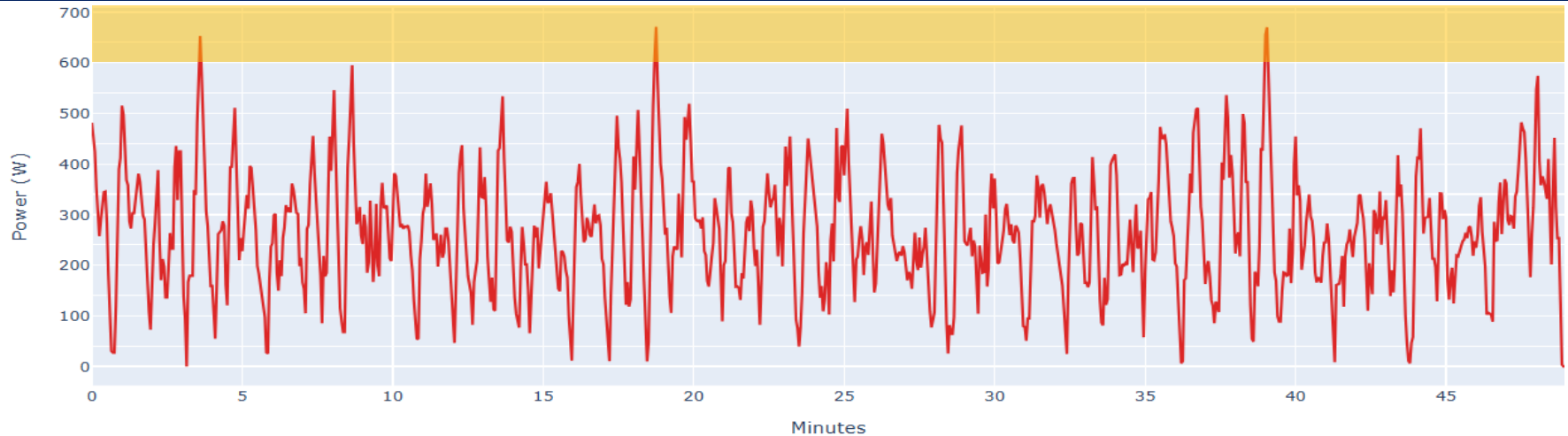
Bike Metrics

Average Power	259 W
Average Power w/o 0's	290 W
Normalized Power	279 W
Variability Index	1.08

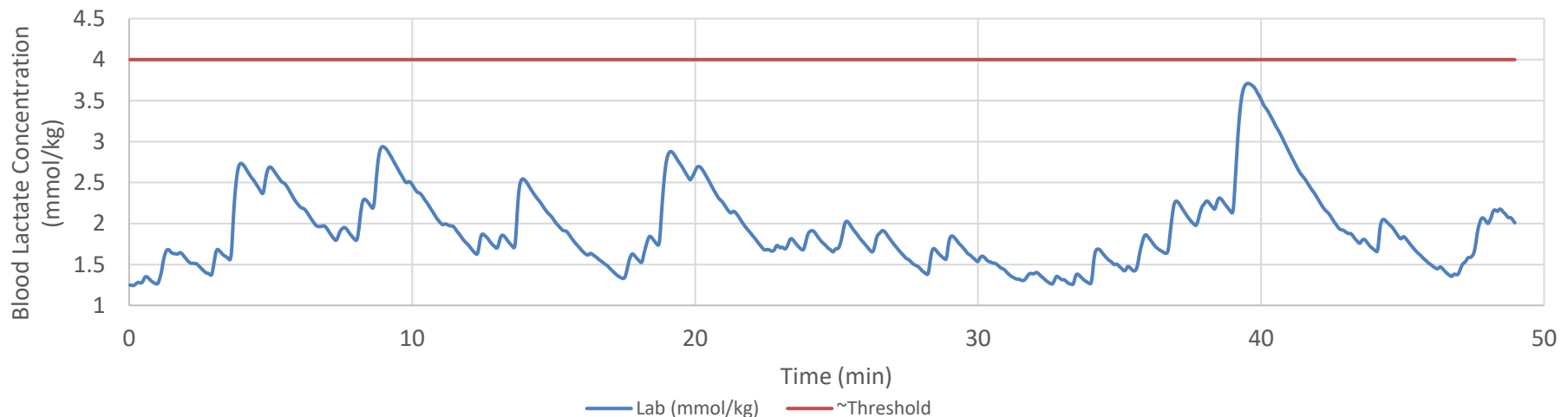
Braxton Legg Yokohama Bike

- As a taller athlete, Braxton is a bit heavier, but also has the highest threshold among the three. Despite an ability to hold higher power, Braxton minimized his surges over 600W to only 3 times during the ride. This kept his lactate below 4 mmol/kg for the whole effort.

Raw Power File



Simulated Blood Lactate Concentration

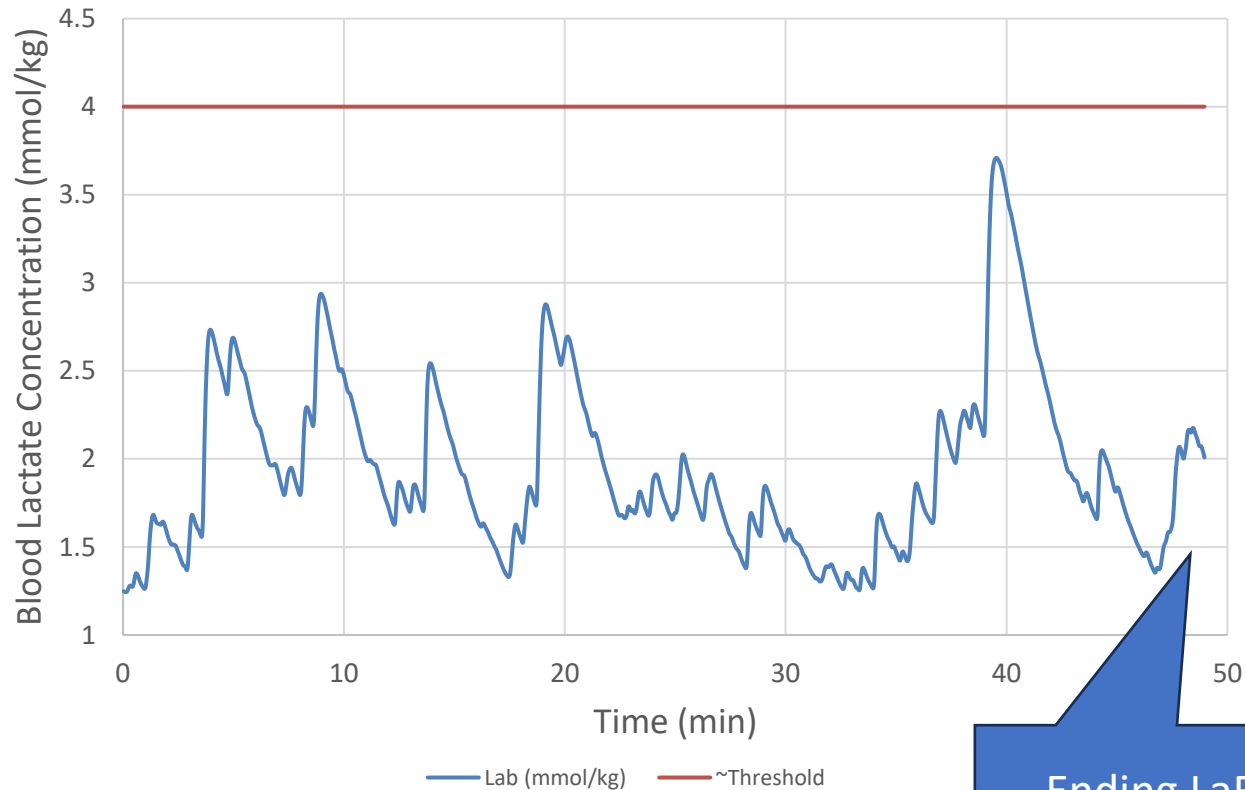


Note: Power file shown with 2s smoothing for better visibility

Braxton Legg Yokohama Bike

- Braxton also rode a mostly consistent effort with a VI of 1.09, keeping any surges to a minimum and riding a large portion of the ride near his maximum lactate clearance zone
- He finished the ride with lactate concentration of 2.06 and went on for his best run performance to date

Simulated Blood Lactate Concentration



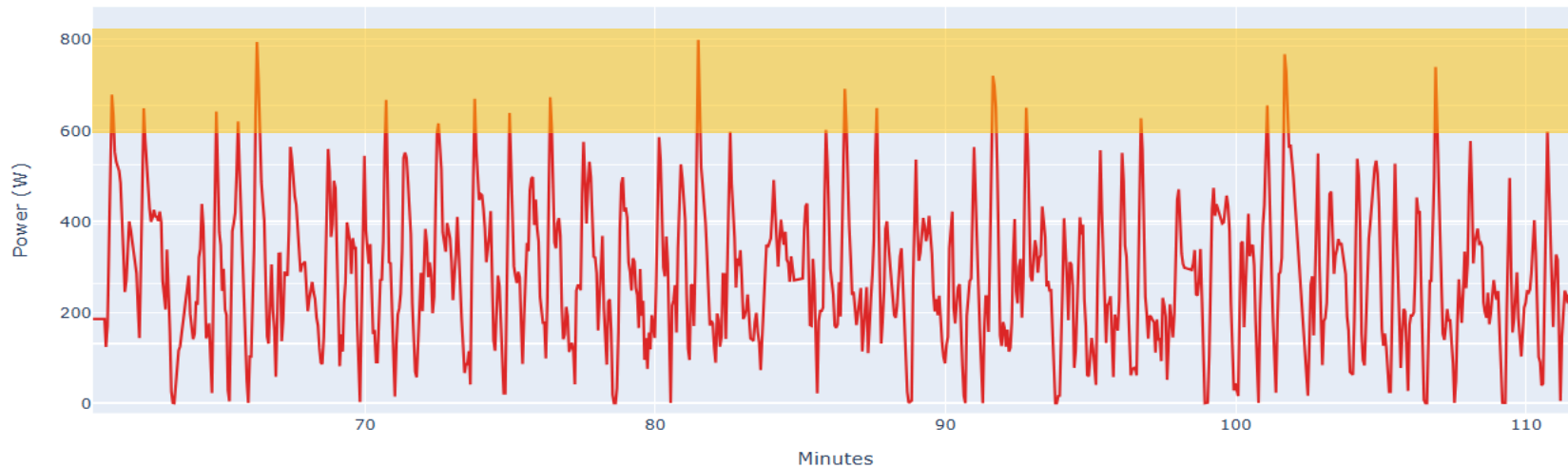
Bike Metrics

Average Power	267 W
Average Power w/o 0's	307 W
Normalized Power	289 W
Variability Index	1.09

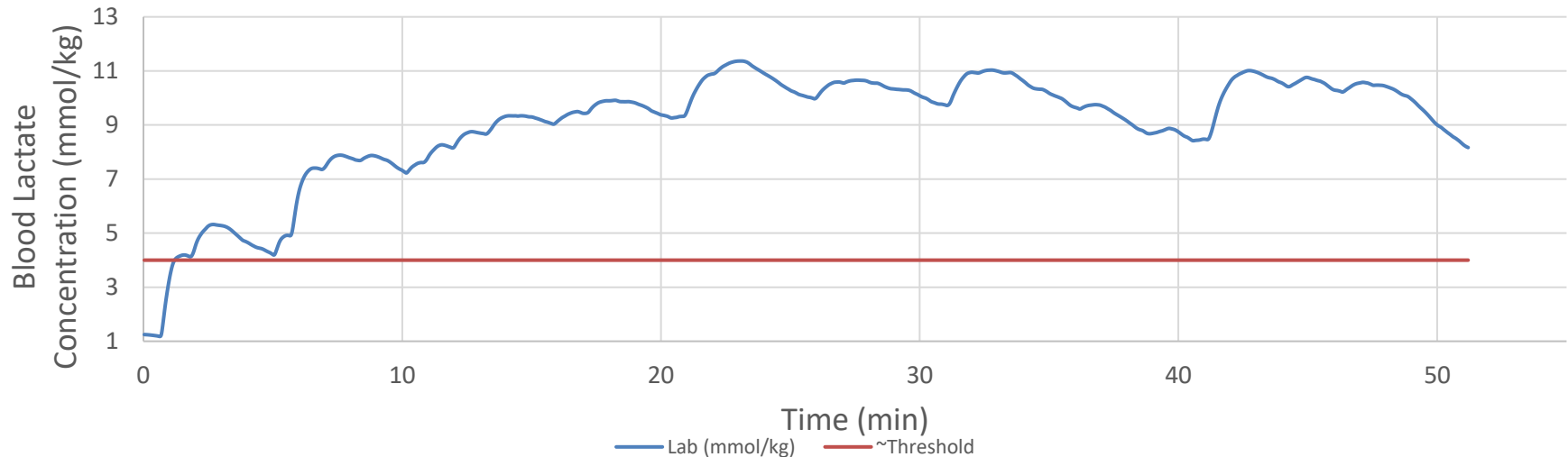
Reese Vannerson Yokohama Bike

- Reese spent significant time over 600 watts with several surges ~800 watts. This caused his lactate to rise significantly above threshold. Additionally, his bike positioning kept him from being able to ride his maximum lactate clearance wattage, keeping lactate high

Raw Power File



Simulated Blood Lactate Concentration

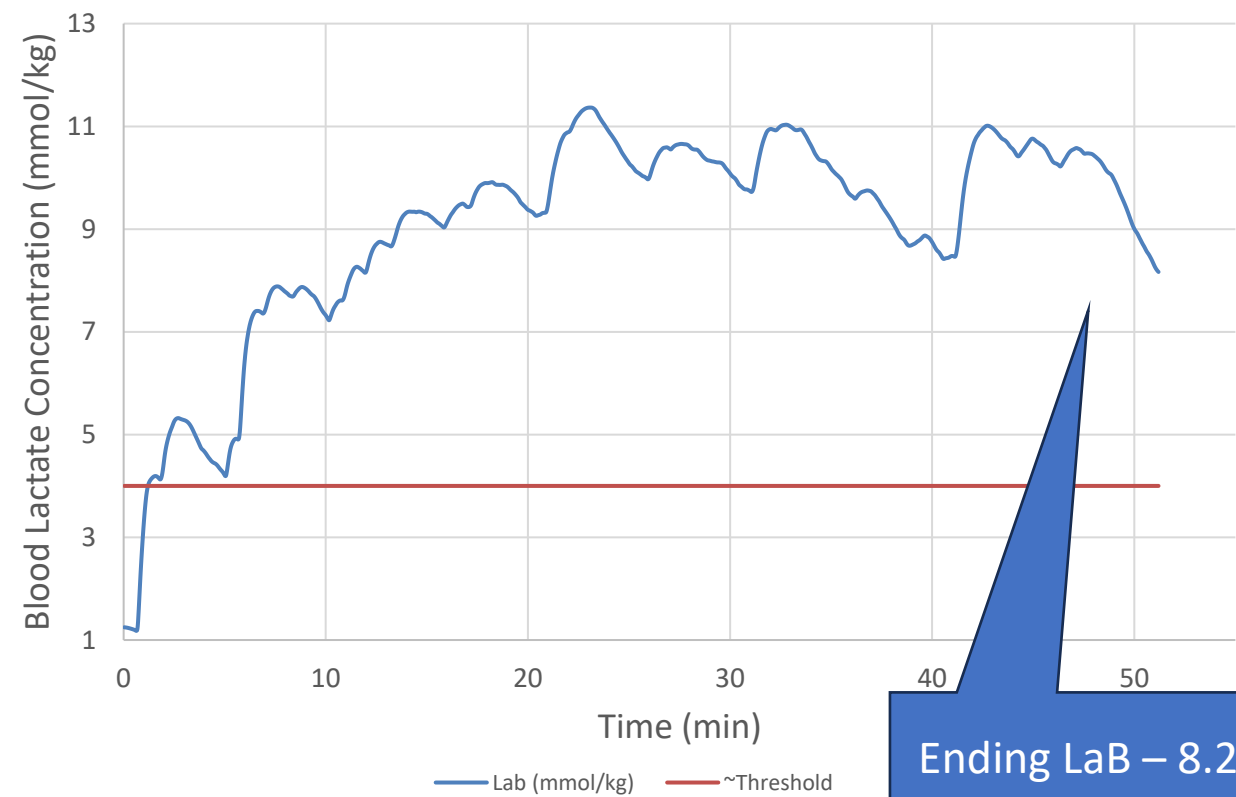


Note: Power file shown with 2s smoothing for better visibility

Reese Vannerson Yokohama Bike

- Despite a similar average power as Blake and Braxton, Reese had a higher Variability Index of 1.14 demonstrating the surges within his ride
- With an ending blood lactate value of 8.21, Reese was significantly over threshold beginning the run

Simulated Blood Lactate Concentration



Bike Metrics

Average Power	274 W
Average Power w/o 0's	314 W
Normalized Power	312 W
Variability Index	1.14